

Solutions to Quiz 2

Question

In each case below provide an example of a sequence of rational numbers satisfying the required property.

- a. A sequence that converges in $|\cdot|_2$ *and* in $|\cdot|_3$ but not in $|\cdot|_\infty$.
- b. A sequence that converges in $|\cdot|_3$ *and* in $|\cdot|_\infty$ but not in $|\cdot|_2$.

Solution:

A sequence (cx^n) goes to 0 in $|\cdot|_*$ if $|x|_* < 1$.

A sequence (cx^n) does not converge in $|\cdot|_*$ if $|x|_* > 1$.

- a. Note that $|6|_2 = 1/2 < 1$ and $|6|_3 = 1/3 < 1$, while $|6|_\infty = 6 > 1$. Thus, the sequence (6^n) converges to 0 in $|\cdot|_2$ and in $|\cdot|_3$, but does not converge in $|\cdot|_\infty$.
- b. Note that $|3/4|_3 = 1/3 < 1$ and $|3/4|_\infty = 3/4 < 1$, while $|3/4|_2 = 4 > 1$. Thus, the sequence $((3/4)^n)$ converges to 0 in $|\cdot|_3$ and in $|\cdot|_\infty$, but does not converge in $|\cdot|_2$.